

Metric-Semantic 3D Reconstruction

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1. Problem Description

Seventeen RGB images of a desktop PC tower (2560 x 1440 pixels) are provided with known 4x4 camera-to-world extrinsic poses and a shared intrinsic matrix. The task is to produce 6-DoF Oriented Bounding Boxes (OBBs) for four specific connectors on the rear I/O panel: ethernet_socket, power_socket, hdmi_socket_left, and usb_socket_top_right. Sixteen frames were part of the original dataset; frame_000465.png was released on the final evaluation day with its pose already included in poses.json.

Each OBB is parameterised by a centre c in R^3 , an extent vector e in R^3 , and a rotation matrix R in $SO(3)$. Grading is by 2D polygonal IoU between the projected OBB and a ground-truth bounding box on held-out test images. The output is a JSON array of four {entity, obb} dictionaries.

2. Literature Review

Multi-view Triangulation

The Direct Linear Transform (DLT) [1] recovers a 3D point from $N \geq 2$ calibrated views by solving a $2N \times 4$ homogeneous linear system via SVD. With known poses (provided here), this is the most direct and accurate path to 3D localisation from pixel correspondences.

Bundle Adjustment

scipy-based bundle adjustment [2] with Huber loss refines 3D points against all observations jointly. Poses are held fixed (given as ground truth); only point positions are optimised. This reduced reprojection RMS from 3.8 px to 0.6 px across 6,321 reconstructed points.

Open-Vocabulary Detection

GroundingDINO [3] couples a DETR-style detector with a BERT text encoder for zero-shot detection from free-form prompts. Multi-prompt aliasing — concatenating alternative phrasings in a single forward pass — improves recall on ambiguous port names.

SAM-Guided Point Lifting

The Segment Anything Model (SAM ViT-H) [4] produces pixel-accurate masks from bounding box prompts. Sparse-cloud points are projected into each frame and filtered to those landing inside the SAM mask, giving far tighter 3D candidate sets than cone-based ray casting for small connectors.

Plane-Snap Regularisation

All rear-panel ports lie on a single physical plane. Projecting any triangulated centre onto that plane — defined by the GT VGA rotation and centre — eliminates the dominant depth error in oblique views, reducing worst-case reprojection from 8-15 px to 1-3 px.

3. System Architecture

The pipeline has two operating modes: a fast deterministic PATH B (correspondence-based, no GPU) used for three entities, and a DINO-driven PATH A used for power_socket which DINO detected reliably across all 17 frames.

```
poses.json + 17 frame images | v [ data_loader.py ] parse c2w matrices, attach K |
+-----+-----+ | | v v [ reconstruction ] [ detection.py ] ORB SfM + BA
GroundingDINO + SAM 6321 3D points mask-guided lifting | | +-----+-----+ | v PATH
A: solve_obb (power_socket via DINO + RANSAC-DLT) PATH B: triangulate from verified
pixel correspondences + plane-snap onto VGA panel | v [ build_answer.py ] schema
validate + write JSON
```

3.1 Coordinate Convention

poses.json stores 4x4 camera-to-world matrices keyed by frame number as a string. The data loader parses each entry as a raw 4x4 list and inverts it to world-to-camera so all downstream geometry uses OpenCV convention $P = K [R | t]$. Image size 2560 x 1440. Intrinsic: $f_x=1477.01$, $f_y=1480.44$, $c_x=1298.25$, $c_y=686.82$. No lens distortion.

3.2 Panel Plane Prior

The instructor supplied the exact VGA OBB as a reference. Its rotation matrix third row is the panel normal in world coordinates. All four required entities lie on the same physical plane, so they share this rotation and their centres are constrained to the plane through the VGA centre with this normal. This structural prior removes two degrees of freedom from each entity's pose problem.

4. Implementation Details

4.1 Sparse Reconstruction (ORB SfM)

ORB with $nfeatures=4000$, $fastThreshold=7$, $scaleFactor=1.2$, $nlevels=8$ on greyscale frames. Cross-checked BFMatcher with Lowe ratio test at 0.75. RANSAC fundamental matrix at 2 px threshold. Union-find track construction. Multi-view DLT triangulation with chirality and 4 px reprojection filters. Final: 6,321 3D points across 17 frames.

4.2 Bundle Adjustment

scipy.optimize.least_squares with $method='trf'$, $loss='huber'$, $f_scale=2.0$. Jacobian sparsity exploited; poses fixed. Reprojection RMS: 3.8 px initial to 0.6 px final over approximately 12,700 observations. Runtime: ~35 seconds on a single CPU core.

4.3 PATH A — DINO + RANSAC-DLT (power_socket)

GroundingDINO-tiny on CUDA with prompts 'power socket. power inlet. c14. ac power.' at $box_threshold=0.20$. Detections found in 17/17 frames. RANSAC-DLT over 300 random 2-view pairs with 8 px inlier threshold and 60 cm sanity prior from VGA centre. Result merged with ray-cone first-surface

estimate, then snapped to panel plane.

4.4 PATH B — Pixel Correspondence Triangulation (ethernet, HDMI, USB)

For entities where DINO produced unreliable or zero detections, pixel centres were manually verified in two or three frames using a dense grid overlay (50 px minor, 200 px major gridlines on full-resolution images). Every 2-view pair is triangulated; the coordinate-wise median is taken for robustness; the result is snapped to the VGA panel plane. Ethernet used three verified frames (449, 461, 465); HDMI and USB used two frames (461, 465).

Entity	Method	Frames used	Max reproj. error
ethernet_socket	PATH B	449, 461, 465	~2 px
power_socket	PATH A	All 17 (DINO)	~2.2 px
hdmi_socket_left	PATH B	461, 465	~3 px
usb_socket_top_right	PATH B	461, 465	~3 px

Table 1. Method and frames used per entity.

4.5 SAM Mask Lifting

For each DINO bounding box, SAM ViT-H is run with the bbox as a box prompt to produce a pixel-accurate segmentation mask. All 6,321 sparse-cloud points are projected into the frame; only those whose projection falls inside the SAM mask are retained as 3D candidates. This replaces cone-based ray casting and significantly reduces background contamination for small connectors with sub-20 mm footprints.

4.6 IEC Standard Extents

Sparse 3D points inside a 16 x 14 mm RJ45 footprint are typically zero or one — insufficient for PCA. Standard mechanical dimensions from IEC connector specifications are used instead: IEC 60603-7 for RJ45 ethernet, IEC 60320 C14 for the power inlet, and measured dimensions for HDMI-A and USB-A 3.0.

4.7 JSON Schema Validation

Before writing, every answer passes through `_validate` in `src/build_answer.py`: top-level type is a list; every entry has `entity` and `obb` keys; `obb.center` and `obb.extent` are length-3; `obb.rotation` is 3x3 with $\det(R)$ approximately +1 and $\|R^T R - I\|$ less than $1e-3$; no NaN or Inf values.

5. Experimental Results

5.1 Reconstruction Quality

Metric	Value
Frames loaded	17 (16 original + frame 465)
Multi-view tracks	6,522
3D points after filtering	6,321
Initial reprojection RMS	3.8 px
Final reprojection RMS	0.6 px (after BA)

Metric	Value
DINO detections	90 across 17 frames

Table 2. Reconstruction and detection metrics.

5.2 OBB Results

Entity	Centre (m)	Extent (m)	Rotation
ethernet_socket	(0.2855, 0.2311, 0.7549)	(0.016, 0.0135, 0.0138)	VGA panel
power_socket	(0.2921, 0.2190, 0.5291)	(0.0278, 0.0198, 0.028)	VGA panel
hdmi_socket_left	triangulated (PATH B)	(0.0145, 0.0055, 0.006)	VGA panel
usb_socket_top_right	triangulated (PATH B)	(0.014, 0.0065, 0.006)	VGA panel

Table 3. OBB results. All entities share the GT VGA rotation matrix since they lie on the same physical I/O panel.

5.3 Visual Validation

All four OBB centres were projected onto frame_000465.png using the supplied pose and the intrinsic matrix. power_socket projects correctly onto the C14 inlet at the base of the tower. ethernet_socket projects onto the RJ45 port in the upper panel cluster. hdmi_socket_left and usb_socket_top_right are verified against the professor's description: HDMI just below the VGA port (leftmost of the two HDMI ports) and USB above the VGA port (rightmost). Annotated verification images are included in docs/obb_debug/.

6. Conclusions

The pipeline successfully produces 6-DoF OBBs for all four required entities on the rear I/O panel of the desktop PC tower. The central insight is that the IoU grading metric is image-space: maximising 2D projection accuracy through careful pixel correspondence, multi-view triangulation, and plane-snap regularisation onto the VGA panel is more effective than maximising 3D metric accuracy via dense reconstruction methods such as 3D Gaussian Splatting.

PATH A (DINO-driven) succeeded for power_socket with 17/17 frame detections and RANSAC-DLT confirmation. For the remaining three entities, PATH B (manual pixel correspondences in frames 461 and 465) was used, producing consistent projections verified visually on frame_000465.png. The two-path design ensures robustness: if automatic detection fails, the correspondence fallback always succeeds given two clear views.

Limitations: (a) DINO failed on ethernet, HDMI, and USB due to ambiguous port appearances and low detection scores; a larger GroundingDINO-SwinB checkpoint would improve recall. (b) With only 2 frames for HDMI and USB triangulation, depth uncertainty is higher than for ethernet (3 frames). (c) Replacing the sparse cloud with a 3DGS scene would provide more surface points for the ray-cone solver.

7. References

[1] R. Hartley and A. Zisserman. Multiple View Geometry in Computer Vision. Cambridge University Press, 2003.

- [2] B. Triggs et al. Bundle Adjustment — A Modern Synthesis. Vision Algorithms: Theory and Practice, 2000.
- [3] S. Liu et al. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. arXiv:2303.05499, 2023.
- [4] A. Kirillov et al. Segment Anything. ICCV, 2023.
- [5] J. L. Schonberger and J.-M. Frahm. Structure-from-Motion Revisited. CVPR, 2016.
- [6] IEC 60603-7. 8-way unshielded connectors. IEC.
- [7] IEC 60320-1. Appliance couplers — General requirements. IEC, 2015.